

Engineering Mathematics Study Notes: Sequences and Series

1. Concept Identification

The provided notes introduce the fundamental building blocks of Real Analysis and Infinite Series Calculus:

- **Sequence:** A function whose domain is the set of positive integers.
 - **Series:** The algebraic sum of the terms of a sequence.
 - **Sequence of Partial Sums (S_n):** A secondary sequence constructed to analyze the convergence behavior of an infinite series.
 - **Monotonic Sequence:** A sequence that consistently moves in a single directional trend—either entirely non-decreasing or entirely non-increasing.
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2. Mathematical Representations

Definition of a Sequence

A sequence is denoted mathematically as:

$$\{u_n\}_{n=1}^{\infty} = \{u_1, u_2, u_3, \dots, u_n, \dots\}$$

Definition of an Infinite Series

The infinite summation of the sequence terms is written as:

$$S_n = \sum_{n=1}^{\infty} u_n = u_1 + u_2 + u_3 + \dots + u_n + \dots + \infty$$

Sequence of Partial Sums

The partial sums are generated recursively:

$$S_1 = u_1$$

$$S_2 = u_1 + u_2$$

$$S_3 = u_1 + u_2 + u_3$$

$$S_n = u_1 + u_2 + \dots + u_n = \sum_{i=1}^n u_i$$

Conditions for Monotonicity

For a sequence to be classified as monotonic increasing:

$$u_{n+1} \geq u_n$$

For a sequence to be classified as monotonic decreasing:

$$u_{n+1} \leq u_n$$

3. The Algorithm (Procedural Recipe)

To analyze an arithmetic progression sequence and determine its general term (u_n or t_n) as shown in the lecture, execute these procedural steps:

Step 1: Identify the Sequence Type

Examine the given list of numbers to determine if the difference between consecutive terms is constant.

$$d = u_{n+1} - u_n$$

Step 2: Extract Key Parameters

Identify the first term (a) and calculate the common difference (d).

$$a = u_1$$

$$d = u_2 - u_1$$

Step 3: Substitute into the General Term Formula

Plug the extracted parameters a and d into the Arithmetic Progression general term formula:

$$t_n = a + (n - 1)d$$

Step 4: Simplify Symbolically

Expand the algebraic expression and collect like terms to arrive at the final simplified formula for the n -th term.

4. Operational Mechanics

Mechanical Example 1: Finding the General Term (t_n)

The whiteboard shows an example sequence: 1, 3, 5, 7, 9, ...

Identify the first term:

$$a = 1$$

Calculate the constant difference between adjacent elements ($3 - 1, 5 - 3, 7 - 5$):

$$d = 2$$

Apply the general formula:

$$t_n = 1 + (n - 1)2$$

Distribute the constant multiplier 2 across the parenthesis:

$$t_n = 1 + 2n - 2$$

Combine the constants ($1 - 2$):

$$t_n = 2n - 1$$

Mechanical Example 2: Understanding Partial Sums

An infinite series adds terms infinitely, which is mechanically impossible to compute directly. Instead, we compute finite chunks called "Partial Sums".

Given the sequence terms generated above ($u_n = 2n - 1$):

$$u_1 = 1, \quad u_2 = 3, \quad u_3 = 5$$

Compute the first partial sum:

$$S_1 = u_1 = 1$$

Compute the second partial sum:

$$S_2 = u_1 + u_2 = 1 + 3 = 4$$

Compute the third partial sum:

$$S_3 = u_1 + u_2 + u_3 = 1 + 3 + 5 = 9$$

The behavior of this new sequence of totals ($\{1, 4, 9, \dots\}$) dictates whether the entire infinite addition approaches a stable finite number or escapes to infinity.

Mechanical Example 3: Verifying Monotonicity

To determine if a sequence is monotonic, compare any arbitrary term u_n to its immediate successor u_{n+1} .

Consider the sequence from the board: 1, 3, 5, 7, $\dots$

$$u_1 = 1, \quad u_2 = 3$$

Evaluate the inequality:

$$3 \geq 1 \implies u_2 \geq u_1$$

Since every subsequent term is greater than or equal to the preceding term ($u_{n+1} \geq u_n$), the sequence is mechanically verified as **monotonic increasing**.

Consider a fractional sequence from the board: $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$

$$u_1 = 1, \quad u_2 = 0.5$$

Evaluate the inequality:

$$0.5 \leq 1 \implies u_2 \leq u_1$$

Since every subsequent term is smaller than or equal to the preceding term ($u_{n+1} \leq u_n$), the sequence is mechanically verified as **monotonic decreasing**.